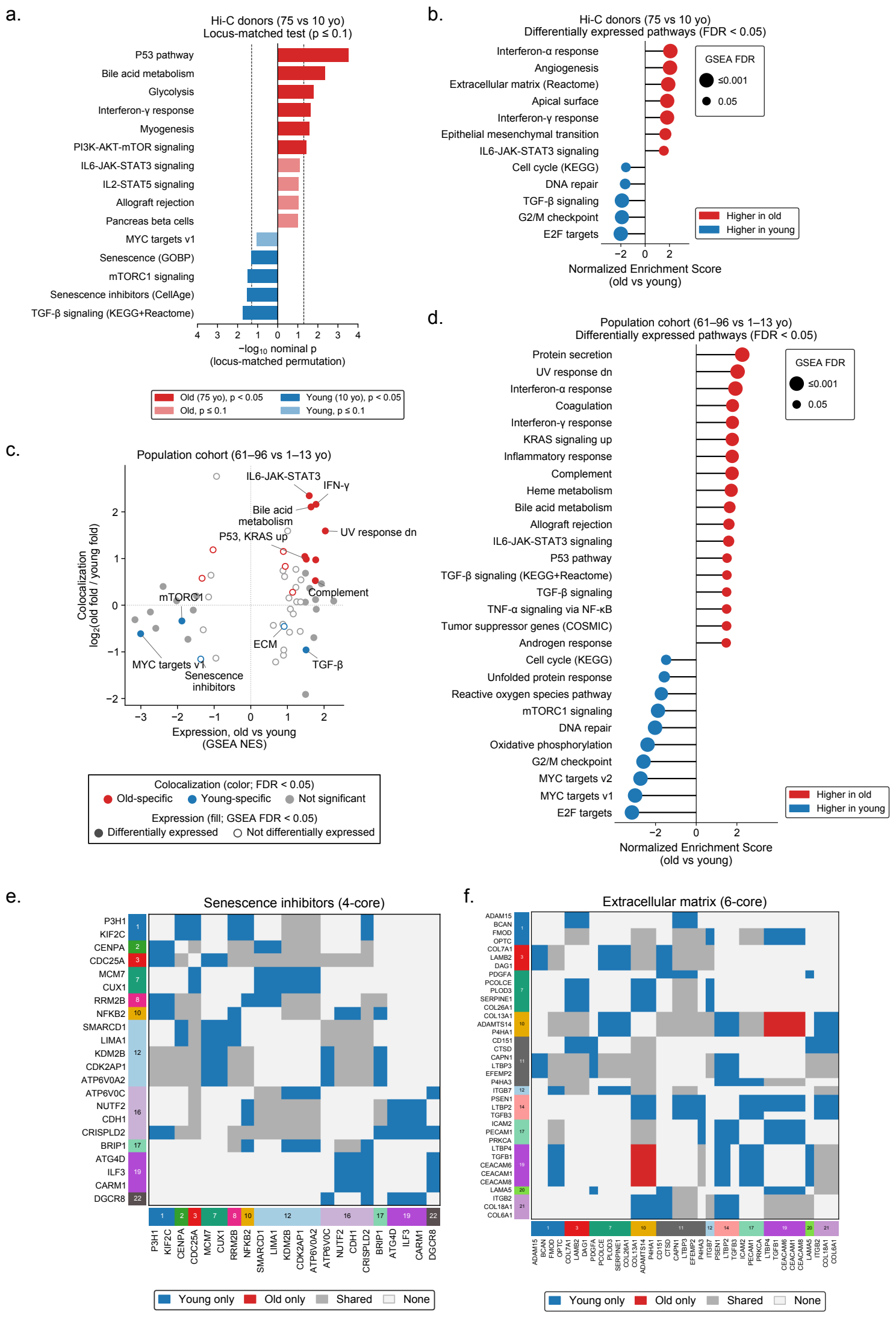

a.
Hi-C donors (75 vs 10 yo)
Locus-matched test (p ≤ 0.1)
P53 pathway
Bile acid metabolism
Glycolysis
Interferon-γ response
Myogenesis
PI3K-AKT-mTOR signaling
IL6-JAK-STAT3 signaling
IL2-STAT5 signaling
Allograft rejection
Pancreas beta cells
MYC targets v1
Senescence (GOBP)
mTORC1 signaling
Senescence inhibitors (CellAge)
TGF-β signaling (KEGG+Reactome)
4 3 2 1 0 1 2 3 4
−log10 nominal p
(locus-matched permutation)
Old (75 yo), p < 0.05
Old, p ≤ 0.1
Young (10 yo), p < 0.05
Young, p ≤ 0.1
b.
Hi-C donors (75 vs 10 yo)
Differentially expressed pathways (FDR < 0.05)
Interferon-α response
Angiogenesis
Extracellular matrix (Reactome)
Apical surface
Interferon-γ response
Epithelial mesenchymal transition
IL6-JAK-STAT3 signaling
Cell cycle (KEGG)
DNA repair
TGF-β signaling
G2/M checkpoint
E2F targets
GSEA FDR
≤0.001
0.05
Higher in old
Higher in young
−2 0 2
Normalized Enrichment Score
(old vs young)
c.
Population cohort (61–96 vs 1–13 yo)
IL6-JAK-STAT3
IFN-γ
Bile acid metabolism
UV response dn
P53, KRAS up
Complement
mTORC1
ECM
TGF-β
MYC targets v1
Senescence inhibitors
Colocalization
log2(old fold / young fold)
Expression, old vs young
(GSEA NES)
Colocalization (color; FDR < 0.05)
Old-specific
Young-specific
Not significant
Expression (fill; GSEA FDR < 0.05)
Differentially expressed
Not differentially expressed
d.
Population cohort (61–96 vs 1–13 yo)
Differentially expressed pathways (FDR < 0.05)
Protein secretion
UV response dn
Interferon-α response
Coagulation
Interferon-γ response
KRAS signaling up
Inflammatory response
Complement
Heme metabolism
Bile acid metabolism
Allograft rejection
IL6-JAK-STAT3 signaling
P53 pathway
TGF-β signaling (KEGG+Reactome)
TGF-β signaling
TNF-α signaling via NF-κB
Tumor suppressor genes (COSMIC)
Androgen response
Cell cycle (KEGG)
Unfolded protein response
Reactive oxygen species pathway
mTORC1 signaling
DNA repair
Oxidative phosphorylation
G2/M checkpoint
MYC targets v2
MYC targets v1
E2F targets
GSEA FDR
≤0.001
0.05
Higher in old
Higher in young
−2 0 2
Normalized Enrichment Score
(old vs young)
e.
Senescence inhibitors (4-core)
P3H1
KIF2C
CENPA
CDC25A
MCM7
CUX1
RRM2B
NFKB2
SMARCD1
LIMA1
KDM2B
CDK2AP1
ATP6V0A2
ATP6V0C
NUTF2
CDH1
CRISPLD2
BRIP1
ATG4D
ILF3
CARM1
DGCR8
Young only
Old only
Shared
None
f.
Extracellular matrix (6-core)
ADAM15
BCAN
FMOD
OPTC
COL7A1
LAMB2
DAG1
PDGFA
PCOLCE
PLOD3
SERPINE1
COL26A1
COL13A1
ADAMTS14
P4HA1
CD151
CTSD
CAPN1
LTBP3
EFEMP2
P4HA3
ITGB7
PSEN1
LTBP2
TGFB3
ICAM2
PECAM1
PRKCA
LTBP4
TGFB1
CEACAM6
CEACAM1
CEACAM8
LAMA5
ITGB2
COL18A1
COL6A1
Young only
Old only
Shared
None